



Introduction to APIs

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What is an API?

An API, or Application Programming Interface, is a way for two software applications to communicate with each other. APIs allow developers to build applications that can interact with other applications, without having to know the underlying implementation of those applications.





APIs are used in a wide variety of applications, including

- Web applications: APIs are used to build web applications that can interact with other web applications, such as social media platforms and e-commerce platforms.
- Mobile applications: APIs are used to build mobile applications that can interact with other mobile applications, as well as with web applications and backend systems.
- Desktop applications: APIs are used to build desktop applications that can interact with other desktop applications, as well as with web applications and backend systems.
- Internet of Things (IoT) devices: APIs are used to build IoT devices that can interact with other IoT devices, as well as with web applications and backend systems.



How do APIs work?

APIs work by providing a set of endpoints that applications can call to exchange data. An endpoint is a URL that represents a specific resource or operation. For example, an API might have an endpoint that allows applications to get a list of users, or an endpoint that allows applications to create a new user.

To use an API, an application will make an HTTP request to an endpoint. The HTTP request will specify the type of operation that the application wants to perform, as well as any data that the application wants to send to the API.

The API will then process the request and return an HTTP response. The HTTP response will contain the data that the application requested, or a status code indicating whether the request was successful.



Endpoint

An endpoint is a URL that represents a specific resource or operation in a REST API. It is the point of entry to the API, and it is where clients send HTTP requests to interact with the API.

Endpoints are typically defined in the API's documentation. The documentation will describe the different endpoints that are available, as well as the data that is required and returned by each endpoint.

For example, a social media platform might have an endpoint that allows clients to get a list of all users, an endpoint that allows clients to create a new user account, and an endpoint that allows clients to update a user's profile information.

When a client wants to interact with the API, it will send an HTTP request to one of the endpoints. The HTTP request will specify the type of operation that the client wants to perform, as well as any data that the client wants to send to the API.



The API will then process the request and return an HTTP response. The HTTP response will contain the data that the client requested, or a status code indicating whether the request was successful.

Endpoints are an important part of REST APIs, as they provide a way for clients to interact with the API in a standardized way. Endpoints make it easy for developers to build applications that can interact with the API, without having to know the underlying implementation of the API.

Here are some examples of endpoints:

- `/users`
- `/users/{id}`
- `/posts`
- `/posts/{id}`
- `/comments`
- `/comments/{id}`

End User with
Browser



Request
→
←
Response

API



Server Back-end
System



Customer

Make the
Order
→
←
Delivery of
order



Waiter

Take the
Order
→
←
Bringing
from Kitchen



Chef



Benefits of using APIs

- Ease of use: APIs make it easy for developers to build applications that can interact with other applications. Developers do not need to know the underlying implementation of the other applications, as long as they know how to use the APIs.
- Flexibility: APIs are flexible and can be used to build a wide variety of applications.
- Scalability: APIs can be scaled to support a large number of users and applications.
- Interoperability: APIs allow applications from different vendors to communicate with each other.



What is a REST API?

A REST API, or Representational State Transfer API, is a way for two computer systems to communicate with each other over the internet. REST APIs are often used to build web services, which are applications that can be accessed through a web browser or mobile app.

REST APIs use a simple and flexible set of rules to define how data is exchanged between the two systems. These rules are based on the HTTP protocol, which is the same protocol that is used to transfer web pages.



Why are REST APIs popular?

REST APIs are popular because they are easy to use and understand. They are also flexible and scalable, which means that they can be used to build a wide variety of web services.

REST APIs are also used by many popular websites and services, such as Google, Facebook, and Amazon. This makes it easy for developers to build applications that can interact with these services.



There are five main HTTP methods that are used in REST APIs

- GET: The GET method is used to retrieve a representation of a resource. For example, you could use the GET method to retrieve a list of all users in a database.
- POST: The POST method is used to create a new resource. For example, you could use the POST method to create a new user account in a database.
- PUT: The PUT method is used to update a resource. For example, you could use the PUT method to update a user's profile information in a database.
- PATCH: The PATCH method is used to partially update a resource. For example, you could use the PATCH method to update a user's email address in a database.
- DELETE: The DELETE method is used to delete a resource. For example, you could use the DELETE method to delete a user account from a database.

C



Create



POST

R



Read



GET

U



Update



PUT

D



Delete



DELETE

edureka!

HTTP
Methods



Get only name and description of task

Request

```
curl -H "Authorization: Bearer <personal_access_token>" "https://app.asana.com/api/1.0/tasks/1224?opt_fields=name,notes"
```

Response

```
HTTP/1.1 200
```

```
{
  "data": {
    "name": "Make a list",
    "notes": "Check it twice!",
    "id": 1224
  }
}
```

HTTP Status Codes

Level 200

200: OK
201: Created
202: Accepted
203: Non-Authoritative
Information
204: No content

Level 400

400: Bad Request
401: Unauthorized
403: Forbidden
404: Not Found
409: Conflict

Level 500

500: Internal Server Error
501: Not Implemented
502: Bad Gateway
503: Service Unavailable
504: Gateway Timeout
599: Network Timeout



1XX	Informational codes	The server acknowledges and is processing the request.
2XX	Success codes	The server successfully received, understood, and processed the request.
3XX	Redirection codes	The server received the request, but there's a redirect to somewhere else (or, in rare cases, some additional action other than a redirect must be completed).
4XX	Client error codes	The server couldn't find (or reach) the page or website. This is an error on the site's side.
5XX	Server error codes	The client made a valid request, but the server failed to complete the request.



A word cloud featuring various international words for "Thank You" in different colors and sizes, centered around a background image of hands holding a globe. The words include: Thank You (large red), Obrigado (grey), Matondo (grey), Kiitos (grey), Mamanana (grey), Maake (grey), Niringrazzjak (grey), Multumesc (grey), Asante (grey), Vinaka (grey), Mercii (grey), Juspaxa (grey), DANK JE (grey), Mochchakkeram (grey), Terma Kasih (grey), Grazie (grey), ASANTE (grey), Raibh Maith Agat (grey), Ua Tsaug Rau Koj (grey), Chokrane (grey), Welalin (grey), Matur Nuwun (grey), Salammat (grey), Arigato (grey), MAAKE (grey), and many others. The background shows two hands holding a small globe, symbolizing global unity.